

SmartHome Hub X100 - Technical Manual

SmartHome Hub X100 – Complete Technical Manual

1. Product Overview

The SmartHome Hub X100 is a centralized smart home controller designed to connect and manage IoT devices including smart lights, thermostats, cameras, motion sensors, smart locks, and voice assistants.

The hub supports:

- Wi-Fi 2.4GHz / 5GHz
- Zigbee 3.0
- Bluetooth Low Energy (BLE)
- Ethernet LAN connection

The system allows users to automate devices using schedules, triggers, and voice commands.

2. Hardware Specifications

Component	Specification
CPU	Quad-Core ARM Cortex A53
RAM	4 GB
Storage	32 GB eMMC
Connectivity	Wi-Fi, Ethernet, Zigbee, BLE
Power Input	12V DC
Operating Temperature 0°C to 45°C	

3. Initial Setup

1. Connect the power adapter.
2. Wait for the LED to turn solid blue.
3. Download the SmartHome Mobile App.
4. Create a user account.
5. Scan the QR code located under the hub.
6. Connect the hub to Wi-Fi.

If setup fails:

- Verify internet connectivity.
 - Restart the router.
 - Hold the reset button for 5 seconds.
-

4. LED Status Indicators

LED Color	Meaning
-----------	---------

Blue Solid	System operational
------------	--------------------

Blue Blinking	Pairing mode
---------------	--------------

Red Solid	System error
-----------	--------------

Red Blinking	Network disconnected
--------------	----------------------

Yellow	Firmware update in progress
--------	-----------------------------

5. User Authentication

The system supports:

- Email/password login
- Google OAuth
- Two-factor authentication (2FA)
- Face recognition through the mobile application

Failed login attempts above 5 trigger temporary account lockout for 15 minutes.

6. Device Pairing

To pair a Zigbee device:

1. Open the mobile app.
2. Navigate to Devices > Add Device.
3. Select Zigbee Device.
4. Put the device in pairing mode.
5. Wait for automatic discovery.

Maximum supported devices:

- 128 Zigbee devices
- 32 BLE devices

- Unlimited cloud devices
-

7. Automation Rules

Examples of automation:

- Turn on lights at sunset.
- Lock doors automatically at midnight.
- Send notifications when motion is detected.
- Turn off air conditioning when windows are open.

Automation execution delay should not exceed 2 seconds under normal network conditions.

8. Firmware Updates

Firmware updates are downloaded automatically.

Manual update process:

1. Open Settings.
2. Navigate to System > Firmware.
3. Click Check for Updates.

4. Download and install.

Do not disconnect power during updates.

9. Common Troubleshooting

Problem: Hub Offline

Possible causes:

- Internet outage
- Router DHCP failure
- Ethernet cable disconnected

Solutions:

- Restart the hub.
 - Reboot the router.
 - Reconfigure network settings.
-

Problem: Device Not Pairing

Possible causes:

- Device too far from hub
- Battery low
- Unsupported firmware version

Solutions:

- Move device closer.
 - Replace batteries.
 - Update firmware.
-

Problem: Mobile App Cannot Connect

Solutions:

- Clear app cache
 - Reinstall application
 - Verify cloud service availability
-

10. Security Recommendations

- Change default passwords immediately.
 - Enable two-factor authentication.
 - Update firmware monthly.
 - Do not expose admin panel to the public internet.
-

11. Factory Reset

To perform a factory reset:

1. Hold reset button for 15 seconds.
2. Wait until LED flashes red three times.
3. Device restarts automatically.

Factory reset deletes:

- Paired devices
 - Automation rules
 - User sessions
 - Local logs
-

12. FAQ

Q: How many devices can the hub manage?

A: Up to 160 local smart devices simultaneously.

Q: Does the hub support Alexa?

A: Yes, Alexa and Google Assistant are fully supported.

Q: Can the hub work offline?

A: Local automations continue working offline, but cloud features require internet access.

Q: What happens during a power outage?

A: Automation resumes automatically once power returns.